

# ZONGYUAN LI

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## Research Interests

Regularity theory of elliptic and parabolic PDEs.

## Employment

|                                                                  |             |
|------------------------------------------------------------------|-------------|
| City University of Hong Kong, Assistant Professor (Tenure-Track) | 2024 –      |
| Rutgers University, Hill Assistant Professor (Non Tenure Track)  | 2020 – 2023 |

## Education

|                                                                |             |
|----------------------------------------------------------------|-------------|
| Brown University, Ph.D. in Applied Mathematics                 | 2015 – 2020 |
| Thesis: Elliptic boundary value problems on irregular domains. |             |
| Advisor: Prof. Hongjie Dong.                                   |             |
| Fudan University, B.Sc. in Mathematics                         | 2010 – 2015 |
| Thesis: Birkhoff normal forms in some PDEs.                    |             |
| Advisor: Prof. Xiaoping Yuan.                                  |             |
| University of Oxford, Part C in Math                           | 2013 – 2014 |
| Disseratation: Study of Keldysh Equation and Related PDEs      |             |
| Thesis supervisor: Prof. Gui-Qiang G. Chen and Dr. Wei Xiang.  |             |
| College Tutor (St. Hugh's College): Dr. Tom Sanders.           |             |

## Long-term Visits

### Visits longer than one week

|                                                      |                              |
|------------------------------------------------------|------------------------------|
| Rutgers University, NJ, USA                          | May – Jun 2026               |
| American Institute of Mathematics, Pasadena, CA, USA | Apr 2024, May 2025, Aug 2026 |
| Institute for Advanced Study, Princeton, NJ, USA     | Jun 2024                     |
| SUNY Binghamton, Vestal, NY, USA                     | Aug – Dec 2023               |

## Grants, Honors & Awards

### Grants

|                                                          |             |
|----------------------------------------------------------|-------------|
| NSFC Young Scientists Fund, #12501275 (RMB 300,000).     | 2026 – 2028 |
| RGC Early Career Scheme (ECS), # 21307225 (HKD 868,287). | 2025 – 2028 |

|                                                          |             |
|----------------------------------------------------------|-------------|
| CityU Start-up Grant, # 7200810 (internal, HKD 300,000). | 2024 – 2026 |
| AMS–Simons Travel Grants (USD 5000).                     | 2021 – 2023 |

### Honors & awards

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|---------------------------------------------------------------------------------------|----------|
| Sigma Xi Award, Brown Applied Math.                                                   | 2020     |
| Outstanding Graduate (ranked 1st in math major), Fudan University.                    | 2015     |
| Bronze Medals in Probability and Statistics, S.-T. Yau College Student Math Contests. | 2013, 14 |
| Bronze Medal in team competition, S.-T. Yau College Student Math Contests.            | 2013     |
| First Prize, Chinese Mathematics Competition (Final Round at Tongji University).      | 2012     |
| First Prize, China National High School Math League.                                  | 2009     |

## Publications

### Papers

- 1 Liouville Theorem with Boundary Conditions from Chern-Gauss-Bonnet Formula. (With B.Z. Chu and Y.Y. Li) Accepted by **Amer. J. Math.** arXiv:2410.16384.
- 2 Asymptotic expansions for harmonic functions at conical boundary points. (With D. Kriventsov) **Rev. Mat. Iberoam.**, 41(3): 837–866, 2025.
- 3 Unique continuation for Robin problems with non-smooth potentials. **J. Funct. Anal.**, 288 (6):Paper No. 110811, 2025.
- 4 On the uniqueness of variable coefficient Schrödinger equations. (With S. Federico and X. Yu) **Commun. Contemp. Math.**, 27 (3), 2025.
- 5 Mixed boundary value problems for parabolic equations in Sobolev spaces with mixed-norms. (With J. Choi and H. Dong) **Calc. Var. Partial Differential Equations.**, 62(1): Paper No. 5, 2023.
- 6 Quantitative unique continuation for Robin boundary value problems on  $C^{1,1}$  domains. (With W. Wang) **Indiana Univ. Math. J.**, 72(4):1429–1460, 2023.
- 7 The Dirichlet-conormal problem for the heat equation with inhomogeneous boundary conditions. (With H. Dong) **Adv. Math.**, 411: Paper No. 108777, 2022.
- 8 Optimal regularity of mixed Dirichlet-conormal boundary value problems for parabolic operators. (With J. Choi and H. Dong) **SIAM J. Math. Anal.**, 54(2):1393-1427, 2022.
- 9 On the  $W_p^2$  estimate for oblique derivative problem in Lipschitz domains. (With H. Dong) **Int. Math. Res. Not. IMRN**, (5):3602-3635, 2022.
- 10 Norm inflation for the Boussinesq system. (With W. Wang) **Discrete Contin. Dyn. Syst. Ser. B**, 26(10):5449-5463, 2021.
- 11 The conormal and Robin boundary value problems in nonsmooth domains satisfying a measure condition. (With H. Dong) **J. Funct. Anal.**, 281(9):Paper No. 109167, 32, 2021.
- 12 The Dirichlet-conormal problem with homogeneous and inhomogeneous boundary conditions. (With H. Dong) **Comm. Partial Differential Equations**, 46(3):470–497, 2021.
- 13 Optimal regularity for a Dirichlet-conormal problem in Reifenberg flat domain. (With J. Choi and H. Dong) **Appl. Math. Optim.**, 83(3):1547-1583, 2021.
- 14 Classical solutions of oblique derivative problem in nonsmooth domains with mean Dini coefficients. (With H. Dong) **Trans. Amer. Math. Soc.**, 373(7) : 4975-4997, 2020.

## Book Chapters

- 1 Asymptotics of harmonic functions in the absence of monotonicity formulas. In *Extended Abstracts 2021/2022*, Trends in Mathematics, vol 3, pages 125 – 131. Birkhäuser, 2024.

## Preprints

- 1 Liouville theorems for conformally invariant fully nonlinear equations. I (With B. Z. Chu and Y. Y. Li) arXiv:2311.07542.
- 2 On the fully nonlinear Yamabe problem with constant boundary mean curvature. I (With B. Z. Chu and Y. Y. Li) arXiv:2410.09683.

## Mentorship

### Ph.D. student

Yuhong XU Expected in 2026 Fall

### Others

Yuhong XU, undergraduate final year project at CityUHK 2025 – 26  
 Cheng Shing CHUA, summer research internship student from NUS 2025  
 Ruixiu LYU, Ms.C. (Financial Mathematics and Statistics) dissertation at CityUHK 2024 – 25

## Teaching

### CityUHK Math

MA 4545 (Undergraduate Differential Geometry) 2025, 2026 Fall  
 MA 2185 (Discrete Math for Computer Sciences) 2025, 2026 Fall  
 MA 3512 (Undergraduate PDE) 2024 Fall  
 MA 3526 (Analysis) 2024, 2025, 2026, 2027 Spring

### Rutgers Math

Math 421 (Calculus for Engineer) 2022 Spring  
 Math 252 (ODE) 2021 Spring, 2022 Fall, 2023 Spring  
 Math 423 (Undergraduate PDE) 2020, 2021, 2022 Fall

### Brown Applied Math (TA)

AM 2640 (Graduate Probability II) 2020 Spring  
 AM 2550 (Graduate Numerical PDE I) 2019 Fall  
 AM 2110 (Graduate Real Analysis I) 2018 Fall  
 AM 33, 34 (ODE I, II) 2016 Fall, 2017 Spring

## Talks

|                                                                                                           |                    |
|-----------------------------------------------------------------------------------------------------------|--------------------|
| PDE seminar, Westlake University.                                                                         | Jun. 2026          |
| Winter workshop on geometric PDEs, Central South University.                                              | Dec. 2025          |
| PDE seminar, Lanzhou University.                                                                          | Dec. 2025          |
| PDE seminar, Duke Kunshan University.                                                                     | Oct. 2025          |
| Symposium on elliptic and parabolic PDEs at Yanqi Lake, Beijing.                                          | Oct. 2025          |
| Short course at Beijing Normal University.                                                                | Aug. 2025          |
| Conformal and CR geometry and their geometric flows, Sanya, China.                                        | Jul. 2025          |
| Recent Advances in Potential Theory and Partial Differential Equations, OIST, Japan.                      | Jul. 2025          |
| International Workshop on Fully Nonlinear Partial Differential Equations, Fudan University.               | Jun. 2025          |
| First Math Workshop of CityU and the University of Edinburgh , Hong Kong.                                 | May 2025           |
| Joint meeting of the NZMS, AustMS and AMS, New Zealand.                                                   | Dec. 2024          |
| Modern Methods in Nonlinear Elliptic and Parabolic PDE, Bulgaria.                                         | May 2024           |
| International Conference on Elliptic and Parabolic Problems: GAETA 2024, Italy.                           | May 2024           |
| PDE seminar, Hong Kong Polytechnic University.                                                            | Mar. and Apr. 2024 |
| AMS sectional meeting, Creighton University.                                                              | Oct. 2023          |
| PDE seminar, Brown University (online).                                                                   | Sep. 2023          |
| Analysis seminar, Temple University.                                                                      | Sep. 2023          |
| Analysis seminar, SUNY Binghamton.                                                                        | Sep. 2023          |
| Colloquium, University of Toronto Scarbrough (online).                                                    | Apr. 2023          |
| Colloquium, City University of Hong Kong (online).                                                        | Apr. 2023          |
| AMS sectional meeting, University of Cincinnati.                                                          | Apr. 2023          |
| Colloquium, University of Nebraska Lincoln.                                                               | Dec. 2022          |
| PDE seminar, Brown University.                                                                            | Dec. 2022          |
| Ghent Methusalem Junior Analysis and PDE Seminar, Ghent University (online).                              | Nov. 2022          |
| Prairie Analysis Seminar, University of Kansas.                                                           | Oct. 2022          |
| AMS sectional meeting, University of Tennessee at Chattanooga.                                            | Oct. 2022          |
| AMS sectional meeting, University of Texas at El Paso.                                                    | Sep. 2022          |
| Analysis seminar, University of Arizona (online).                                                         | Mar. 2022          |
| International Conference on Partial Differential Equations Related to Material Science, Beijing (online). | May 2021           |
| Learning Seminar on PDE and Applications, Rutgers University (online).                                    | Mar. and Apr. 2021 |
| Virtual Analysis and PDE Seminar (VAPS), hosted by Purdue University (online).                            | Jan. 2021          |
| New postdoc special colloquium, Rutgers University (online).                                              | Oct. 2020          |
| Learning Seminar on PDE and Applications, Rutgers University (online).                                    | Aug. 2020          |
| AMS sectional meeting, Tufts University (canceled due to COVID-19).                                       | Mar. 2020          |
| Brown-BU-UMassAmherst PDE and Dynamics seminar.                                                           | Nov. 2019          |

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|------------------------------------------------------------------------|-----------|
| PDE seminar, Brown University.                                         | Nov. 2019 |
| 2019 Northeast Analysis Network Conference, University of Connecticut. | Sep. 2019 |
| PDE seminar, Brown University.                                         | Oct. 2018 |

## Services

### Conferences organized

|                                                                                                                                                                                             |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Co-organizer of the special session <i>Recent Developments in Regularity Theory for PDEs</i> , The 15th AIMS conference, Athens, Greek                                                      | Jul 2026 (scheduled) |
| Co-organizer of the special session <i>Recent Advances in Elliptic PDEs and Their Applications</i> . JMM 2026, Washington, DC., USA                                                         | Jan 2026             |
| Co-organizer of the special session <i>Recent Developments in PDEs and Related Areas</i> , JMM 2025, Seattle, USA                                                                           | Jan 2025             |
| Co-organizer of the special session <i>New trends in elliptic and parabolic PDEs</i> , The 14th AIMS Conference, Abu Dhabi, UAE                                                             | Dec 2024             |
| Co-organizer of the special session <i>Dynamics and Regularity of PDEs</i> , JMM 2024, San Francisco, USA                                                                                   | Jan 2024             |
| Co-organizer of the special session <i>Topics in PDEs and harmonic analysis</i> , AMS Fall Eastern Sectional Meeting, UMass Amherst, USA                                                    | Oct 2022             |
| Co-organizer of the special session <i>Regularity Theory for Linear and Nonlinear PDEs</i> , AMS Spring Western Sectional Meeting (online, formerly at San Francisco State University, USA) | May 2021             |

### Reviewed journal articles

*Adv. Math.*, *Adv. Nonlinear Anal.*, *Ann. Appl. Math.*, *Appl. Math. Lett.*,  
*Calc. Var. Partial Differential Equations*, *Commun. Pure Appl. Anal.*,  
*Discrete Contin. Dyn. Syst. Ser. B*, *J. Differential Equations*, *J. Math. Pures Appl.*,  
*Int. Math. Res. Not. IMRN*, *NoDEA Nonlinear Differential Equations Appl.*,  
*Nonlinearity*, *Nonlinear Anal.*, *Trans. Amer. Math. Soc.*